

# SRS ⇌ BDD Traceability Matrix

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**Sources.** Requirements: `docs/SRS_FACIS_DCS.pdf` (all 225 bracket-defined `[DCS-...]` IDs; the `[DCS-FR-UC-XX-Y]` sections are the SRS's own use-case→requirement linkage and are treated as mapping metadata, not requirements). Coverage: `features/**/*.feature` (behave suite run on every CI push, kind-in-docker).

**2026-07-22 wave note.** Two DCS instances now negotiate a contract end to end ( `full-vertical-2dcs.spec.ts` : offer → negotiation ping-pong with a PDF exchanged per adjustment → settle → dual signing → scoped audit, every workflow action driven through the real UI). Federation follows the prefer-inbound rule — the peer's PDF is authoritative and stored verbatim because it carries provenance we cannot reproduce, regeneration is verify-only — and inbound counter-offers are authorised by counterparty rather than by local RBAC. The audit trail's global tamper evidence moved from a per-event chain to Merkle checkpoints (ADR-16), the contract format collapsed onto one typed `dcx:Placeholder` node (ADR-15) in expanded JSON-LD (ADR-14), and machine callers became SRS System Users with one OAuth2 client per class. Two dispositions in this wave are refusals rather than features: `Sys. Contract Signer` holds no signing scope (ADR-17, eIDAS Art. 3(9)/26 — a signatory is a natural person under sole control, and a legal person's instrument is a seal we do not implement), and external anchoring of the checkpoint chain is wired but not in force, since the ORCE flow polling the head ends at a debug node. New pack this wave: `12/system_user_classes`.

**Claim correction (2026-07-22).** An assessment of `testWallet` against the ARF and the EUDI reference libraries, fetching both verbatim, found the ARF citations used elsewhere in this repo were wrong: Topic 16 is QES-scoped throughout and has no AES track, `QES_05` is remote-QES enrolment rather than the flow list (that is `QES_06`), and CSC API 2.0 was `QES_07`, now `Empty`. Only "QES\_08 = PAdES" holds. Since the SRS descopes QES, **ARF Topic 16 conformance is not a claim this project may make**, and neither is sole control/SCAL2 — `testWallet`'s keys are shared files by design. The same assessment found four verified divergences from the reference wallet that `testWallet` currently accommodates, so a green CI run is not evidence of wallet interoperability.

**2026-07-17 wave note.** First fully green combined CI run: behave (312 scenarios) and the Playwright suite (11 specs) pass in the same kind-stack job (run 29541989685 @ 27c0a1be). Since the 2026-07-16 note the E2E layer grew the Semantic Hub management specs (`e2e/semantic-hub-publish.spec.ts`) and surfaced a fifth UI defect (shacl-form fragments one subject across several `@id-sharing` node objects; merged before emission) plus one product concurrency bug: the async PDF/C2PA pipeline's cache write bumped `contracts.updated_at` and tripped optimistic concurrency for racing editors — fixed so `updated_at` moves only on non-cache column changes (migration 20260717, caught by the C2PA chain scenario).

**2026-07-16 wave note.** A Playwright E2E layer now covers the browser UI (frontend/ClientApp/e2e, run in CI after behave against the same kind stack): real OID4VP sessions per test, API-seeded fixtures, 9/9 specs green. It surfaced and fixed four real UI defects (Pinia boot-order crash; the `odrl:Set-era` is `OdrISet` predicate that blocked rendering of every canonical contract; a crash on `domainField-less RequirementFields`; shacl-form's `rdf:type-vs-@type` serialization that kept ODRL typed clauses from ever becoming rules through the browser). Rows below marked "e2e/" cite that suite. Same wave: `contracts/templates` carry dereferenceable resource IRIs (`@id = {DCS_PUBLIC_URL}/contract/{key}`, resolve routes added), policy sets follow the Offer→Agreement lifecycle sealed at first signature, and the CI run at 920cf253 reduced open BDD failures

to two, both fixed (renewal IRI step; zero-finding contracts now appear in /pac/audit as explicitly compliant).

**2026-07-14 wave note.** The rows citing packs 22/multi\_signer and 23/semantic\_hub, the archive annotation/full-text scenarios (07), the JAdES provenance scenarios (17), the target-acknowledgement/KPI scenarios (05), and the signature view/compliance scenarios (04) were implemented in this wave: all scenarios bind (behave dry-run) and every backend unit test is green, but the wave has not yet had its first full CI run — statuses here reflect the executable evidence as written.

**Method.** Every requirement gets exactly one disposition. Scenario references name the feature pack (by its `features/` directory number) plus a short scenario descriptor; most scenarios also carry the requirement ID as a behave tag ( `@DCS-...` ), so `grep -r <ID> features/` finds the executable evidence.

**Harness scope (applies to all  rows).** The suite is a black-box HTTP harness against the deployed service. Browser-UI behavior, TLS termination, platform hardening, and process/documentation requirements are verified by review/ops, not BDD — the SRS itself lists "Review of Documentation" as the verification method for most of them. This is a documented decision (see the `@skip Signature-Manager-UI` scenario in `features/22_real_signing_vertical` ), not a coverage hole.

| Status                                                                                                 | Meaning                                                                       | Count      |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------|
|  Covered              | scenario(s) or Playwright specs assert the requirement end-to-end             | 156        |
|  In progress        | being implemented                                                             | 0          |
|  Partial            | core behavior asserted; named residue not (yet) provable                      | 42         |
|  Not BDD-verifiable | infrastructure/process requirement — verified outside the automated harnesses | 24         |
|  Deviation          | capability not implemented in the product; recorded deviation                 | 3          |
|                                                                                                        | <b>Total</b>                                                                  | <b>225</b> |

### 3.2.1 Template Repository (DCS-FR-TR-...)

| ID           | Requirement                                          | Status       | Evidence / disposition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-FR-TR-01 | Machine-Readable Format                              | ✓<br>Covered | Templates stored/retrieved as JSON-LD — 02/create_template, 02/template_identity; editor state is the JSON-LD doc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| DCS-FR-TR-02 | Multi-Tiered Contract Template Management            | ✓<br>Covered | 20/hierarchy invariant scenarios (tagged @DCS-FR-TR-02): parent refs, child-enumeration rejection, cycle rejection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| DCS-FR-TR-03 | Semantic Hub for Schema Storage                      | ✓<br>Covered | Semantic Hub built (23/semantic_hub): versioned JSON-LD context + SHACL shape storage seeded with the FACIS v1 profile, public resolution, Template-Manager register/rollback (UC-02-08), every produced document anchored via resolvable standard-vocabulary anchors (@context hub URL, sh:shapesGraph, dcterms:conformsTo), and hub-prefix redefinition rejected at template creation. Hub management UI asserted: e2e/semantic-hub.spec.ts (dashboard lists every registered artifact from the live inventory; clause catalog served with labeled, shape-backed entries) and e2e/semantic-hub-publish.spec.ts (an operator publishes a brand-new shapes entry through the UI and it resolves immediately on the public route — the Gaia-X case; register + activate a new version of an existing entry). ADR-8: enforcement ( <code>AuditContractContent</code> ) reads its SHACL shapes/validation profile from the hub's active (or, for revalidation, pinned-per-document) version, hub-only (no disk fallback) — 23/semantic_hub "Activating a stricter SHACL shapes version..." proves activate/rollback actually changes what gets enforced, and that already-produced contracts stay pinned. ADR-9: the enforcement engine is goRDFlib, a conformant SHACL-core processor verified against the W3C SHACL/SHACL-1.2 suites (388/388, pinned commit recorded in the ADR) — real <code>sh:datatype / sh:minInclusive / sh:pattern / sh:node / sh:nodeKind</code> constraints, not a hand-rolled subset matcher; <code>internal/base/validation/contractcontentaudit_test.go</code> <code>TestAuditContractContentSHACLRejectsWrongDatatype</code> is the unit-level xsd:integer-rejection proof. |
| DCS-FR-TR-04 | Machine-Readable and Human-Readable Template Linking | ✓<br>Covered | MR→HR derivation proven via template PDF export + verify (02/template_integrity_audit); bidirectional <i>link</i> metadata not modeled beyond same-DID pairing. Phase 3 (ADR-10) partially addresses the machine-readable half for clauses specifically: typed clause instances (dcs:PaymentClause etc.) are generated from and validated against the same Semantic Hub SHACL shapes (GET /semantic/clauses, 23/semantic_hub "The clause catalog is seeded..."), so a clause's authored form and its enforcement share one source of truth — <code>TestAuditContractContentValidatesTypedClauses</code> proves server-side enforcement; the frontend palette ( <code>TypedClausePalette.vue</code> ) is manual/UI-review evidence, consistent with the existing DCS-IR-TR partial-row convention. Now fully covered: every machine rule must carry dcs:prose referencing its human-readable clause (Go gate + <code>Odrl*ProseShape</code> SHACL, 18 structure scenario), and e2e/template-typed-clause.spec.ts proves the UI end:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| ID           | Requirement                                       | Status               | Evidence / disposition                                                                                                                                                                                                                                                                                        |
|--------------|---------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                                                   |                      | a hub-palette typed clause becomes a prose-backed rule whose dcs:prose dereferences to a document block.                                                                                                                                                                                                      |
| DCS-FR-TR-05 | Template Version Control                          | ✓ Covered            | Template versions/approvals tracked; template audit-log scenario (02, @DCS-FR-TR-21/TR-05). retrieve_history_by_id exists.                                                                                                                                                                                    |
| DCS-FR-TR-06 | Role-Based Access Control for Template Repository | ✓ Covered            | RBAC negative scenarios: 02/create, 02/update, 02/archive, 02/workflow 'Unauthorized role cannot ...' + 01 pack 401 sweep.                                                                                                                                                                                    |
| DCS-FR-TR-07 | Compliance & Legal Validation                     | ● Partial            | Approval gate before usability proven (02/template_workflow + contract create requires REGISTERED). Domain-specific regulatory rule packs beyond ODRL/structural validation are not modeled.                                                                                                                  |
| DCS-FR-TR-08 | )                                                 | 📄 Not BDD-verifiable | SRS formatting artifact: bracketed ID sits inside the §3.1.1 Template Builder UI narrative (review-task generation + unique ID). Substance covered by 02/template_workflow (submit→review task) and 02/template_identity (unique DID).                                                                        |
| DCS-FR-TR-09 | Template Provenance and Versioning                | ✓ Covered            | Registration seals each version's provenance as a signed W3C VC (JSON-LD, ecdsa-rdfc-2019): creator/reviewer/approver/registrar claims, content hash, previous-credential linkage; served by GET /template/provenance/{did} (02/template_provenance). Provenance also travels in template bundle export (20). |
| DCS-FR-TR-10 | Searchable Metadata & Categorization              | ✓ Covered            | 02/search_templates: by name, description, details; RBAC negative.                                                                                                                                                                                                                                            |
| DCS-FR-TR-11 | Template UUID / DID Assignment                    | ✓ Covered            | 02/template_identity: UUID on creation, retrieve by DID.                                                                                                                                                                                                                                                      |
| DCS-FR-TR-12 | Template Customization                            | ✓ Covered            | SHOULD — 02/generate_contract: contract generated from approved template with populated terms.                                                                                                                                                                                                                |
| DCS-FR-TR-13 | Template Creation                                 | ✓ Covered            | 02/create_template + register step (/template/register), RBAC negative.                                                                                                                                                                                                                                       |
| DCS-FR-TR-14 | Template Submission for Approval                  | ✓ Covered            | 02/template_workflow 'Submit template for review'.                                                                                                                                                                                                                                                            |

| ID           | Requirement                                                            | Status       | Evidence / disposition                                                                                                                                                                                                                                                                                 |
|--------------|------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-FR-TR-15 | Template Approval Process                                              | ✓<br>Covered | 02/template_workflow approve/reject/resubmit set (7 scenarios).                                                                                                                                                                                                                                        |
| DCS-FR-TR-16 | Template Update Management                                             | ✓<br>Covered | 02/update_template (creator, reviewer variants) with version continuity.                                                                                                                                                                                                                               |
| DCS-FR-TR-17 | Template Retirement and Deprecation                                    | ✓<br>Covered | 02/template_archive 'Deprecate an active template' + cannot-delete-deprecated guard.                                                                                                                                                                                                                   |
| DCS-FR-TR-18 | Template Deletion                                                      | ✓<br>Covered | 02/template_archive delete scenarios incl. RBAC negative.                                                                                                                                                                                                                                              |
| DCS-FR-TR-19 | Template Retrieval                                                     | ✓<br>Covered | 02/generate_contract + template_identity retrieve-by-DID (/template/retrieve).                                                                                                                                                                                                                         |
| DCS-FR-TR-20 | Template Compliance and Integrity Verification                         | ✓<br>Covered | /template/verify scenario (02, tagged @DCS-FR-TR-20).                                                                                                                                                                                                                                                  |
| DCS-FR-TR-21 | Audit Logs for Template Changes                                        | ✓<br>Covered | Template audit-log scenario (02, tagged @DCS-FR-TR-21).                                                                                                                                                                                                                                                |
| DCS-FR-TR-22 | Notification System for Template Updates                               | ✓<br>Covered | Webhook platform (/orce): subscribable template.updated/template.registered events fan out to registered receivers with the template DID in the payload; delivery log with acknowledgement (GET /deliveries). Verified end-to-end against the ORCE monitoring flow (02/template_update_notifications). |
| DCS-FR-TR-23 | Structural Dependency Mapping The Template Repository MUST allow Te... | ✓<br>Covered | 20 hierarchy dependency enforcement + export refusal on missing component (tagged @DCS-FR-TR-26/@DCS-FR-PACM-06).                                                                                                                                                                                      |
| DCS-FR-TR-24 | Structural Export in Unified Format                                    | ✓<br>Covered | 20 template bundle export (tagged @DCS-FR-TR-24).                                                                                                                                                                                                                                                      |

| ID           | Requirement                                     | Status                                                                                    | Evidence / disposition                                                                                                                                                                                                                                                                                           |
|--------------|-------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-FR-TR-25 | Multi-Contract Template Builder                 |  Covered | Visual builder is a frontend concern (HTTP-only harness; see the 22 UI-gap precedent). Backing APIs covered via 20 hierarchy/bundle + 02 CRUD. UI now asserted: e2e/template-typed-clause.spec.ts drives the visual builder end-to-end (type selection, block palette, shacl-form, save) against the live stack. |
| DCS-FR-TR-26 | Logical Validation of Structural Dependencies   |  Covered | 20 'Export is refused with a findings list when a referenced component is missing' (tagged).                                                                                                                                                                                                                     |
| DCS-FR-TR-27 | Contract Type Classification                    |  Partial | Multi/single-party structure expressed via responsible-party DIDs and hierarchy; a dedicated contract-type classification facet for filtering is not modeled.                                                                                                                                                    |
| DCS-FR-TR-28 | Template Management Dashboard (see Section 3.1) |  Covered | Dashboard UI; backing APIs (search/status/workflow) covered by 02 pack. UI now asserted: e2e/dashboards.spec.ts (template dashboard lists registered templates).                                                                                                                                                 |

### 3.2.2 Contract Workflow Engine (DCS-FR-CWE-...)

| ID            | Requirement                                                | Status       | Evidence / disposition                                                                                                                                                                                                                                                         |
|---------------|------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-FR-CWE-01 | Multi-Party Contract Management                            | ✓<br>Covered | 17/two_instance: offer with instance-B negotiator+approver, cross-instance OFFERED/APPROVED replication.                                                                                                                                                                       |
| DCS-FR-CWE-02 | Hierarchical Contract Structures                           | ✓<br>Covered | 20 hierarchy scenarios: single-parent model, cycle rejection, frame-contract child listing.                                                                                                                                                                                    |
| DCS-FR-CWE-03 | Contract Assembling                                        | ✓<br>Covered | 'Assemble contract from reusable clauses' (03/contract_creation).                                                                                                                                                                                                              |
| DCS-FR-CWE-04 | Machine-Readable & Human-Readable Contract Synchronization | ✓<br>Covered | 03/format_review MR/HR hash scenarios + 08 verify endpoint + 22 dual-hash binding scenario (all tagged @DCS-FR-CWE-04).                                                                                                                                                        |
| DCS-FR-CWE-05 | Secure Human-Readable Contract Viewer                      | ✓<br>Covered | Tamper-evidence of the served HR view proven via verify + tamper seams (03/format_review). Viewer UI itself out of harness scope. Viewer UI now asserted: e2e/dashboards.spec.ts renders the human-readable document from the machine-readable JSON-LD (Contract Content tab). |
| DCS-FR-CWE-06 | Event-Driven Contract Execution                            | ✓<br>Covered | 05 auto-deployment on signing completion; 15 re-approval flow (tagged @DCS-FR-CWE-06); events logged (08).                                                                                                                                                                     |
| DCS-FR-CWE-07 | Role-Based Access Control                                  | ✓<br>Covered | Role-guard negatives across 03/05/07/08/22; credential-based roles via OIDC (01).                                                                                                                                                                                              |
| DCS-FR-CWE-08 | Version Control                                            | ✓<br>Covered | Version history via /contract/retrieve_history_by_id — 'Track version history during negotiation' (03).                                                                                                                                                                        |
| DCS-FR-CWE-09 | SLA & Compliance Monitoring                                | ✓<br>Covered | 05 KPI ingestion + SLA-violation flag scenarios (tagged @DCS-FR-CWE-09).                                                                                                                                                                                                       |
| DCS-FR-CWE-10 | Contract Expiry                                            | ✓<br>Covered | 19 expiry-cron scenario: 'expired' lifecycle banner after cron fires.                                                                                                                                                                                                          |

| ID            | Requirement                    | Status       | Evidence / disposition                                                                                                                                                                                                                                                     |
|---------------|--------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-FR-CWE-11 | Contract Renewal               | ✓<br>Covered | POST /contract/renew; renewal keeps refs to prior DID/version (06 pack).                                                                                                                                                                                                   |
| DCS-FR-CWE-12 | Termination Handling           | ✓<br>Covered | 06/contract_termination: terminate with reason, double-termination guard (tagged @DCS-FR-CWE-11/12).                                                                                                                                                                       |
| DCS-FR-CWE-13 | Contract Creation              | ✓<br>Covered | 03/contract_creation create-from-template, editable versioned draft.                                                                                                                                                                                                       |
| DCS-FR-CWE-14 | Contract Submission for Review | ✓<br>Covered | 03 state-machine submit→review→approve chain; 'Submit contract for review after negotiation'.                                                                                                                                                                              |
| DCS-FR-CWE-15 | Contract Review and Approval   | ✓<br>Covered | 03/contract_approval approve/reject/initiate; partial-quorum enforcement proven with two DISTINCT approver peers in 17/two_instance's approval-quorum scenario: one approval leaves the contract REVIEWED, the second flips it APPROVED with both peer decisions recorded. |
| DCS-FR-CWE-16 | Contract Initiation            | ✓<br>Covered | 'Contract transitions to signing phase upon approval' (03); sign-after-approve proven in 22.                                                                                                                                                                               |
| DCS-FR-CWE-17 | Contract Review                | ●<br>Partial | Redlining + version compare via history endpoint (03); automated missing-field checks via structural validation (20 hierarchy rejections). Side-by-side diff is a UI concern.                                                                                              |
| DCS-FR-CWE-18 | Contract Negotiation           | ✓<br>Covered | 03/contract_negotiation: comments, redlines (green); decision rounds + negotiation log covered.                                                                                                                                                                            |
| DCS-FR-CWE-19 | Contract Signing               | ✓<br>Covered | 22 end-to-end AES signing with ceremony, status tracked (ceremony + e2e scenarios).                                                                                                                                                                                        |
| DCS-FR-CWE-20 | Store Contract in Archive      | ✓<br>Covered | 05 archive-at-SIGNED scenario (tagged @DCS-FR-CWE-20).                                                                                                                                                                                                                     |
| DCS-FR-       | Retrieve Contract from Archive | ✓<br>Covered | 07 archive retrieve/search with RBAC.                                                                                                                                                                                                                                      |



| ID            | Requirement                            | Status                                                                                      | Evidence / disposition                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------|----------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CWE-21        |                                        |                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| DCS-FR-CWE-22 | Contract Renewal Management            |  Covered   | Renewal workflow endpoint (see CWE-11).                                                                                                                                                                                                                                                                                                                                                                             |
| DCS-FR-CWE-23 | Contract Termination                   |  Covered   | 06 termination via API, removed from active flows (state TERMINATED).                                                                                                                                                                                                                                                                                                                                               |
| DCS-FR-CWE-24 | Contract Management Dashboard          |  Covered   | Dashboard UI; backing search/status APIs covered (03 state-filtered search, 07). UI now asserted: e2e/dashboards.spec.ts (contract dashboard lists contracts with lifecycle state).                                                                                                                                                                                                                                 |
| DCS-FR-CWE-25 | Contract Review and Approval Interface |  Partial   | Approval API surface covered (03/contract_approval); dedicated reviewer UI out of harness scope.                                                                                                                                                                                                                                                                                                                    |
| DCS-FR-CWE-26 | Contract Signing Interface             |  Partial  | Signing API + ceremony covered (22); browser signing UI documented out-of-scope (22 @skip UI scenario).                                                                                                                                                                                                                                                                                                             |
| DCS-FR-CWE-27 | Contract Tracking and Status Overview  |  Covered | 03 state-machine state-filtered search; status history via retrieve_history_by_id (approval routing scenario).                                                                                                                                                                                                                                                                                                      |
| DCS-FR-CWE-28 | Automated Contract Interaction via API |  Covered | 12/contract_lifecycle_via_api: full lifecycle + queryable history via API. Machine callers now have real identities: one OAuth2 client_credentials client per SRS §2.4 Table 5 System User class, with roles fixed by deployment configuration because such a token carries no verifiable role claims (ADR-16) — 12/system_user_classes obtains real Hydra tokens and checks what each class may and may not reach. |
| DCS-FR-CWE-29 | Multi-Contract Visualization           |  Covered | 20 parent_id search filter + frame-contract detail (tagged @DCS-FR-CWE-29).                                                                                                                                                                                                                                                                                                                                         |
| DCS-FR-CWE-30 | Contract Package Bundling              |  Covered | 20 bundle-export scenarios: ZIP members, parent-chain refs, manifest hashes (tagged @DCS-FR-CWE-30).                                                                                                                                                                                                                                                                                                                |
| DCS-FR-       | Contract Performance Tracking          |  Covered | 05 ACTIVE-in-live-list + KPI-on-detail scenarios (tagged @DCS-FR-CWE-31).                                                                                                                                                                                                                                                                                                                                           |

| ID     | Requirement | Status | Evidence / disposition |
|--------|-------------|--------|------------------------|
| CWE-31 |             |        |                        |

### 3.2.3 Signature Management (DCS-FR-SM-...)

| ID           | Requirement                                                            | Status  | Evidence / disposition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------|------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-FR-SM-01 | Level of Assurance Flexibility for Simple Electronic Signature, Adv... | Partial | AES level proven end-to-end (22 e2e scenario); credential_type honored (apply-fields scenario). QES requires a qualified TSP/QSCD — unavailable in hermetic env; deviation note for QES execution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DCS-FR-SM-02 | Support for PAdES, JAdES, and CAdES Signatures                         | Covered | PAdES B-T proven (22) with ETSI.CAdES.detached CMS container (CAdES); JAdES baseline-B implemented for the machine-readable contract in the DCS-to-DCS flow (17 provenance + tamper-negative scenarios; internal/base/jades unit tests).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| DCS-FR-SM-03 | Signing Identity and PoA Authorization Credentials                     | Partial | Signer identity credential (PID SD-JWT VC) verified before signing (22 ceremony-gate, webhook/PID-embedding, and verify cross-check scenarios). PoA (dc+sd-jwt, vct urn:dc:poa:v1) is presented at LOGIN and mapped into the Hydra session — every authenticated call is PoA-gated; issuer chain-walk to trust anchor stays open (recorded deviation; SRS TBD-B acknowledges XFSC PCM unavailability). A machine caller cannot sign at all: eIDAS Art. 3(9) makes a signatory a natural person and Art. 26 requires sole control, so the System Contract Signer class holds no signing scope (ADR-17) — 12/system_user_classes asserts the refusal at ceremony start, prepare and submit, and that the same client may still verify. |
| DCS-FR-SM-04 | Counterparty Authorization and PoA Credential Chain Verification       | Partial | Credential status/revocation is checked on every verification (status-list check in each verify path) — a revoked PoA blocks the login that gates signing. Chain-walk to a trust anchor remains roadmap (recorded deviation).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| DCS-FR-SM-05 | Integration with Signing Identity and PoA Verifiable Credentials       | Covered | W3C-compliant SD-JWT VC + KB-JWT presented, verified, embedded verbatim under the PAdES signature (22 verbatim-embedding + verify cross-check scenarios).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| DCS-FR-SM-06 | Wallet for Identity, PoA Credential Management, and Signing            | Partial | Wallet protocol surface (OID4VP presentation, headless) proven (22 webhook + headless-ceremony scenarios); a real end-user wallet app is outside the harness.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| DCS-FR-SM-07 | Multi-Signature and Role-Based Signing Flows                           | Covered | 22/multi_signer: one ceremony + one sequential PAdES signature per declared field, all-ceremonies-before-first-signature evidence embedding, ceremony-gate and double-signing negatives, deploy gate until every field is signed; role gating via 22 ceremony role-denial.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| DCS-FR-      | Persisted Contract                                                     | Covered | 22 ContractSigningSummaryCredential issued + embedded; PDF/A-3 attachment under signature (tagged @DCS-FR-SM-08). Phase 4 (ADR-9):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| ID           | Requirement                                                        | Status       | Evidence / disposition                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|--------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SM-08        | Signing Summary with Verifiable Credential and P...                |              | the credential now also carries schema_version/validation_report_hash — the Semantic Hub SHACL version the contract validated against at signing time and a stable hash of the findings (validation.SHACLEvidence) — and signature/validate re-runs pinned-version validation and cross-checks the hash for drift (crossCheckSHACLDrift, backend/internal/signingmanagement/query/validate.go), unit-tested via TestSHACLEvidenceIsStableAndDetectsDrift. |
| DCS-FR-SM-09 | Secure Human-Readable Contract Viewer                              | ✓<br>Covered | Same as CWE-05: tamper-evidence of served content proven; viewer UI out of harness. Viewer UI now asserted alongside CWE-05: e2e/dashboards.spec.ts.                                                                                                                                                                                                                                                                                                      |
| DCS-FR-SM-10 | Proof of Contract Execution                                        | ✓<br>Covered | 05 TSA-timestamped execution receipt appended to archive (tagged @DCS-FR-SM-10).                                                                                                                                                                                                                                                                                                                                                                          |
| DCS-FR-SM-11 | Linked Machine-Readable and Human-Readable Signatures              | ✓<br>Covered | 22 signature record binds PDF hash + JSON-LD content hash.                                                                                                                                                                                                                                                                                                                                                                                                |
| DCS-FR-SM-12 | Contract Deployment Trigger                                        | ✓<br>Covered | 05 deploy-trigger scenarios incl. auto-trigger on signing (tagged @DCS-FR-SM-12).                                                                                                                                                                                                                                                                                                                                                                         |
| DCS-FR-SM-13 | Signature Workflow Process                                         | ✓<br>Covered | 22 ceremony orchestration and lifecycle statuses.                                                                                                                                                                                                                                                                                                                                                                                                         |
| DCS-FR-SM-14 | Signature Request from Signer                                      | ✓<br>Covered | 22 POST /signature/request + status polling (tagged @FR-SM-14).                                                                                                                                                                                                                                                                                                                                                                                           |
| DCS-FR-SM-15 | Contract Retrieval for Signing                                     | ✓<br>Covered | Signed-PDF retrieval with validation exercised throughout 22 (IPFS-CID persisted artifact).                                                                                                                                                                                                                                                                                                                                                               |
| DCS-FR-SM-16 | Apply Digital Signature (via Cloud PCM or OCM Signer API Endpoint) | ✓<br>Covered | 22 real PAdES via HSM path (tagged @DCS-FR-SM-16).                                                                                                                                                                                                                                                                                                                                                                                                        |
| DCS-FR-      | Multi-Signer Support                                               | ✓<br>Covered | 22/multi_signer end-to-end: two DISTINCT signer identities recorded independently per field (signature view assertion), sequential application on                                                                                                                                                                                                                                                                                                         |

| ID           | Requirement                              | Status               | Evidence / disposition                                                                                                                                                                                                                                                  |
|--------------|------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SM-17        |                                          |                      | signed bytes (mechanics also unit-proven by pdf-core TestPADESSecondSignatureProbe); parallel signing stays a documented change request.                                                                                                                                |
| DCS-FR-SM-18 | Signature Validation                     | ✓<br>Covered         | Signature validate endpoint scenario (04/signature_validation, tagged @DCS-FR-SM-18).                                                                                                                                                                                   |
| DCS-FR-SM-19 | Audit Log for Signatures                 | ✓<br>Covered         | Signature audit-log scenario (04, tagged @DCS-FR-SM-19).                                                                                                                                                                                                                |
| DCS-FR-SM-20 | Signature Revocation                     | ✓<br>Covered         | 15 revocation → REVOKED + re-approval path (tagged @DCS-FR-SM-20).                                                                                                                                                                                                      |
| DCS-FR-SM-21 | Signature Compliance Verification        | ✓<br>Covered         | Signature compliance endpoint scenario (04, tagged @DCS-FR-SM-21).                                                                                                                                                                                                      |
| DCS-FR-SM-22 | Signature Dashboard for Contract Signers | ✓<br>Covered         | Signer dashboard UI; backing status API covered (22 status polling). UI now asserted: e2e/dashboards.spec.ts (signing dashboard renders for the signer role).                                                                                                           |
| DCS-FR-SM-23 | Signing Interface                        | 📄 Not BDD-verifiable | Browser signing UI + biometrics: documented out of harness (22 @skip UI scenario records the decision).                                                                                                                                                                 |
| DCS-FR-SM-24 | Signature Status Tracking                | ✓<br>Covered         | 22 ceremony-status-progression scenario.                                                                                                                                                                                                                                |
| DCS-FR-SM-25 | Automated Signature Processing API       | ✓<br>Covered         | 22 fully headless API-driven ceremony (tagged @FR-SM-25).                                                                                                                                                                                                               |
| DCS-FR-SM-26 | Signature Compliance Viewer              | 🟡<br>Partial         | GET /signature/view serves the viewer's full data set — per-signature signer identity, field, credential class, status, timestamps, container format + integrity findings (04 view scenarios incl. RBAC negative); the Vue viewer itself stays out of the HTTP harness. |
| DCS-FR-SM-27 | Support for PDF/A Format                 | ✓<br>Covered         | 04/signature_validation asserts PDF/A-3 identification on the exported SIGNED PDF bytes (pdfaid:part=3, conformance=A, ISO 19005-3) plus the contract.jsonld associated file (AFRelationship /Source); full veraPDF-class validation remains an external check.         |

### 3.2.4 Contract Storage & Archive (DCS-FR-CSA-...)

| ID            | Requirement                             | Status                                                                                      | Evidence / disposition                                                                                                                                                                                                                                                                                                                                                     |
|---------------|-----------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-FR-CSA-01 | Tamper-Proof Contract Storage           |  Covered   | Merkle-checkpointed, TSA-anchored audit trail (ADR-16) + tamper scenarios (03/format_review tampered-PDF) prove tamper-evidence. A submitted PDF may add a signature but never redefine the contract: the submit path compares the embedded JSON-LD against the prepared document, deliberately the opposite of the federation rule where an inbound PDF is authoritative. |
| DCS-FR-CSA-02 | Role-Based Access Control               |  Covered   | 07 role-outside-archive-scope denied; access audited (20 export audit-log scenario).                                                                                                                                                                                                                                                                                       |
| DCS-FR-CSA-03 | Proof-of-Existence                      |  Covered   | TSA timestamp + IPFS anchoring per event (05 TSA receipt, 08 audit anchoring).                                                                                                                                                                                                                                                                                             |
| DCS-FR-CSA-04 | Contract Expiry & Renewal Tracking      |  Partial   | Expiry detection + banner proven (19). Configurable-threshold alert notifications not modeled — deviation note.                                                                                                                                                                                                                                                            |
| DCS-FR-CSA-05 | Hierarchical Contract Storage           |  Covered | 20 sibling-isolation + party-scoped-bundle scenarios: hierarchy preserved and scoped in archive/bundles.                                                                                                                                                                                                                                                                   |
| DCS-FR-CSA-06 | Machine-Readable Contract Storage       |  Covered | JSON-LD stored + exported alongside PDF (20 bundle members; 05 deploy-payload shape); sync validated pre-archive via verify.                                                                                                                                                                                                                                               |
| DCS-FR-CSA-07 | Automated Compliance Checks             |  Partial | ODRL/structural gates block non-compliant contracts before they can reach SIGNED/archive (18 approve/sign gates, 20 export refusal); a distinct archive-time re-check is not separate from the workflow gate.                                                                                                                                                              |
| DCS-FR-CSA-08 | Store Signed Contract in Archive        |  Covered | 05 archive-at-SIGNED: archive entry exactly on SIGNED with evidence.                                                                                                                                                                                                                                                                                                       |
| DCS-FR-CSA-09 | Generate and Assign Contract Identifier |  Covered | Contract DIDs assigned at creation and used across workflows (03, 12, 17).                                                                                                                                                                                                                                                                                                 |
| DCS-FR-CSA-10 | Index Contract Metadata                 |  Covered | 07 state-filtered archive search; archive metadata view (contracts_archive_metadata).                                                                                                                                                                                                                                                                                      |

| ID            | Requirement                                    | Status               | Evidence / disposition                                                                                                                                                                |
|---------------|------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-FR-CSA-11 | Create Contract Summary and Tags               | ✓ Covered            | 07 annotation scenarios: manual + metadata-generated summaries, tag assignment, tag-filtered search (inclusion and exclusion), RBAC negative, ANNOTATE_ARCHIVED_CONTRACT audit event. |
| DCS-FR-CSA-12 | Retrieve Contract from Archive                 | ✓ Covered            | 07 archive retrieval with RBAC + audit (20 export audit-log).                                                                                                                         |
| DCS-FR-CSA-13 | Search Contracts                               | ✓ Covered            | Metadata/state search (07) plus full-text content search over the whole contract JSON-LD (stored tsvector, GIN-indexed) — 07 full-text scenario with positive and negative queries.   |
| DCS-FR-CSA-14 | Contract Expiration Handling                   | ✓ Covered            | 19 expired banner + expiry cron; expired contracts excluded from active workflows.                                                                                                    |
| DCS-FR-CSA-15 | Contract Renewal and Extension                 | ✓ Covered            | Renewal contract linked to archived original (06, tagged @DCS-FR-CSA-15).                                                                                                             |
| DCS-FR-CSA-16 | Contract Termination                           | ✓ Covered            | 06 termination with reason recorded; terminated contracts remain retrievable read-only (07 search by state).                                                                          |
| DCS-FR-CSA-17 | Contract Deletion                              | ✓ Covered            | Archive delete scenarios (07, tagged @DCS-FR-CSA-17) incl. audit logging.                                                                                                             |
| DCS-FR-CSA-18 | Audit Log for Contract Storage and Retrieval   | ✓ Covered            | 20 export RBAC + audit-entry scenarios (tagged @DCS-FR-CSA-18); archive audit endpoint covered (07).                                                                                  |
| DCS-FR-CSA-19 | Compliance Verification for Archived Contracts | ● Partial            | Audit entries retrievable per component (07/08); automated compliance flagging of archived entries beyond workflow gates not modeled.                                                 |
| DCS-FR-CSA-20 | Automated Contract Monitoring and Alerts       | ● Partial            | pac/monitor continuous monitoring (08); configurable UI/email alert delivery not modeled — deviation note.                                                                            |
| DCS-FR-CSA-21 | Contract Archive Dashboard                     | 📄 Not BDD-verifiable | Dashboard UI; backing stats/search APIs covered (07).                                                                                                                                 |

| ID            | Requirement                                        | Status                                                                                               | Evidence / disposition                                                                                                                                                        |
|---------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-FR-CSA-22 | Contract Search Interface                          |  Not BDD-verifiable | Search UI; backing API covered (07 archive search).                                                                                                                           |
| DCS-FR-CSA-23 | Contract Expiration and Renewal Management UI      |  Not BDD-verifiable | Expiry/renewal UI; backing expiry + renewal APIs covered (19, 06).                                                                                                            |
| DCS-FR-CSA-24 | Contract Compliance and Audit Viewer               |  Covered            | Audit viewer UI; backing pac/report + archive audit APIs covered (08, 07). UI now asserted: e2e/dashboards.spec.ts (audit workstation renders scoped audits for the auditor). |
| DCS-FR-CSA-25 | Contract Processing API                            |  Covered            | Archive store/retrieve/search/delete APIs with authz + audit (07 pack, 20 export audit-log).                                                                                  |
| DCS-FR-CSA-26 | Archive Multi-Party Contract Component Assignments |  Covered            | 20 sibling isolation across instances + party-scoped bundle content (tagged @DCS-FR-CSA-26).                                                                                  |



### 3.2.5 Process Audit & Compliance (DCS-FR-PACM-...)

| ID             | Requirement                                                 | Status                                                                                         | Evidence / disposition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-FR-PACM-01 | Tamper-Proof Audit Trail for Contract Lifecycle             | <br>Covered   | 08 process audit incl. create event; exportable via /pac/report. Global tamper evidence is now a Merkle checkpoint per anchoring batch, each root chained to its predecessor and timestamped once (ADR-16), replacing the per-event global chain that serialised every event behind one TSA round-trip and stalled the trail when a single event failed; per-resource hash chains are unchanged. Leaves are blinded so a published proof reveals nothing, and 12/system_user_classes asserts a checkpoint head carrying a root and leaking no entry data. An unanchorable event is dead-lettered rather than retried silently, i.e. a gap is visible. NOT yet in force: external anchoring — the ORCE flow polling the head ends at a debug node, so the log is tamper-evident to us but not provable against the operator. |
| DCS-FR-PACM-02 | Compliance Monitoring and Risk Detection                    | <br>Covered   | 08 continuous monitoring; risk-during-approval scenario (03/contract_approval).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| DCS-FR-PACM-03 | Automated Regulatory and Policy Compliance Checks           | <br>Covered   | 18 ODRL gates on approve+sign; /pac/monitor sweep flags MISSING_APPROVAL risks on approval-pending contracts and anchors each as PAC_COMPLIANCE_RISK per contract (03/contract_approval monitoring scenario, 08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| DCS-FR-PACM-04 | Role-Based Access Control for Audit Logs                    | <br>Partial | 08 non-auditor denied. Per-access justification recording not modeled — noted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| DCS-FR-PACM-05 | Contract Non-Compliance Investigation and Reporting         | <br>Covered | 08 incident-report scenario; monitor + report link findings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| DCS-FR-PACM-06 | Structural Integrity Validation for Multi-Contract Packages | <br>Covered | 20 export-refusal structural-integrity scenario (tagged @DCS-FR-PACM-06).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DCS-FR-PACM-07 | Compliance Reporting by Contract Component and Party        | <br>Partial | Scoped audit/report per component (08 scoped audit/report scenarios). Per-party/per-clause segmentation not modeled — noted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### 3.1.4 Communications Interfaces (DCS-IR-CI-...)

| ID           | Requirement                     | Status                                                                                               | Evidence / disposition                                                                                                                                               |
|--------------|---------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-IR-CI-01 | HTTPS/TLS 1.3 Transport         |  Not BDD-verifiable | TLS termination is deployment config (prod ingress); BDD kind env intentionally runs plaintext behind Traefik. Verify via deployment values review.                  |
| DCS-IR-CI-02 | REST/JSON API Conventions       |  Covered            | All suite traffic is REST/JSON; PDFs served as application/pdf (03 format review HR export).                                                                         |
| DCS-IR-CI-03 | Browser Access over HTTPS       |  Not BDD-verifiable | HTTPS for UI = same deployment concern as CI-01.                                                                                                                     |
| DCS-IR-CI-04 | OAuth2/OIDC Flows               |  Covered            | 01 pack: OIDC login/refresh/logout/introspection paths incl. expired-credential rejection.                                                                           |
| DCS-IR-CI-05 | OpenID Discovery & JWKS         |  Covered            | Token validation against Hydra discovery/JWKS exercised by every authenticated scenario; expired-JWT scenario pins issuer handling (01).                             |
| DCS-IR-CI-06 | OpenID4VC/VP Bindings           |  Partial           | OID4VP presentation flow proven headlessly (22 webhook/PID-embedding scenarios). OID4VCI issuance is the wallet/issuer side, outside DCS runtime — noted.            |
| DCS-IR-CI-07 | Orchestration Webhooks          |  Covered          | 05 ORCE Node-RED flow round-trip incl. hash verification + ack.                                                                                                      |
| DCS-IR-CI-08 | DSS Remote Signing over HTTPS   |  Partial          | Internal signing endpoints (c2paSign/padesSign) fill the DSS role in-cluster (21 internal-signing scenario); external DSS/TSP over HTTPS not reachable hermetically. |
| DCS-IR-CI-09 | Revocation List Synchronization |  Partial          | CRL revocation flip proven (21); the ≤5-minute propagation bound is not timed in-suite.                                                                              |
| DCS-IR-CI-10 | PACM Audit Event Transport      |  Covered          | 08 pack uses /pac/audit + /pac/report over HTTPS JSON (transport per CI-01 in prod).                                                                                 |

### 3.1.3 Software Interfaces (DCS-IR-SI-...)

| ID           | Requirement                                             | Status       | Evidence / disposition                                                                                                                                                                                                                                                                                                                                                                                |
|--------------|---------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-IR-SI-01 | Template Catalogue Integration                          | ✓<br>Covered | Template catalogue endpoints scenarios (02/template_catalogue, tagged @DCS-IR-SI-01).                                                                                                                                                                                                                                                                                                                 |
| DCS-IR-SI-02 | Workflow Orchestration (Node-RED) Integration           | ✓<br>Covered | 05 shipped ORCE contract-target flow round-trip (tagged @DCS-IR-SI-02).                                                                                                                                                                                                                                                                                                                               |
| DCS-IR-SI-03 | Platform Authentication & Authorization Integration     | ✓<br>Covered | 01 pack — all components enforce OAuth2/OIDC.                                                                                                                                                                                                                                                                                                                                                         |
| DCS-IR-SI-04 | Wallet & TSP Signing Integration                        | ●<br>Partial | OID4VP + remote-signing seam proven via headless ceremony + HSM signing (22); real TSP integration out of hermetic scope.                                                                                                                                                                                                                                                                             |
| DCS-IR-SI-05 | External Target System API Integration                  | ✓<br>Covered | 05 external target deploy API scenarios incl. shared-secret callback (tagged @DCS-IR-SI-05).                                                                                                                                                                                                                                                                                                          |
| DCS-IR-SI-06 | Counterparty DCS Information Endpoint                   | ✓<br>Covered | 17 get_sync/post_sync + GetServiceDID: policy-gated peer information exchange (untrusted peer rejected).                                                                                                                                                                                                                                                                                              |
| DCS-IR-SI-07 | OpenID Provider Discovery & JWKS Consumption            | ✓<br>Covered | Hydra discovery/JWKS consumption (see CI-05).                                                                                                                                                                                                                                                                                                                                                         |
| DCS-IR-SI-08 | OpenID4VP Login & Access Control                        | ✓<br>Covered | OID4VP login is the only authentication path: Hydra login+consent is accepted solely after trust-anchored presentation verification ( <code>auth_login.go</code> PresentationCallback → <code>oid4vp.Verify</code> → <code>AcceptLoginAndConsent</code> ), and every authenticated scenario performs the full headless VP login. JAR ES256-signed (21); PID presentation re-verified at signing (22). |
| DCS-IR-SI-09 | Credential Status & Revocation Service                  | ✓<br>Covered | 21 CRL/status-list revocation flip.                                                                                                                                                                                                                                                                                                                                                                   |
| DCS-IR-SI-10 | Digital Signature Service (DSS) Authorization & Signing | ●<br>Partial | DSS-shaped authorize+sign+timestamp path via internal signing + TSA (22 timestamp scenario); external DSS not hermetic.                                                                                                                                                                                                                                                                               |
| DCS-IR-SI-11 | Relational Database Access                              | ✓<br>Covered | PostgreSQL with versioned migrations exercised by the entire suite (backend/migrations/sql).                                                                                                                                                                                                                                                                                                          |
| DCS-IR-SI-   | Crypto Provider & DID/VC Operations                     | ✓<br>Covered | 21 HSM-backed DID/VC/C2PA operations (Crypto-Provider role).                                                                                                                                                                                                                                                                                                                                          |

| ID | Requirement | Status | Evidence / disposition |
|----|-------------|--------|------------------------|
| 12 |             |        |                        |

### 3.1.1 UI — Template Repository (DCS-IR-TR-...)

| ID           | Requirement                                                            | Status                                                                                         | Evidence / disposition                                                                                                                                                                                                                                                    |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-IR-TR-01 | Template Builder MUST allow Template Creator to create new contract... | <br>Covered   | API: 02 create/update template. Builder UI now asserted: e2e/template-typed-clause.spec.ts creates a brand-new template through the visual builder (/ui/templates/new → hub typed-clause palette → shacl-form → save) and verifies the emitted /template/create envelope. |
| DCS-IR-TR-02 | Template Builder MUST allow searching and retrieving existing templ... | <br>Partial   | API: 02 search/retrieve. Template listing UI asserted (e2e/dashboards.spec.ts); in-UI search interaction unasserted.                                                                                                                                                      |
| DCS-IR-TR-03 | Template Review MUST allow Reviewers to retrieve, verify, update, a... | <br>Partial   | API: 02 workflow review steps. UI out of harness.                                                                                                                                                                                                                         |
| DCS-IR-TR-04 | Template Review MUST support forwarding a verified template to appr... | <br>Partial | API: 02 approve/reject/resubmit transitions. UI out of harness.                                                                                                                                                                                                           |
| DCS-IR-TR-05 | Template Approval MUST allow Approvers to retrieve, approve, reject... | <br>Partial | API: 02 approval set. UI out of harness.                                                                                                                                                                                                                                  |
| DCS-IR-TR-06 | Template Approval MUST ensure that only validated templates enter t... | <br>Partial | API: only REGISTERED templates usable in contract create (03 steps). UI out of harness.                                                                                                                                                                                   |
| DCS-IR-TR-07 | Template Management Dashboard MUST allow Managers to register, arch... | <br>Partial | API: 02 register/archive/update/search + audit. UI out of harness.                                                                                                                                                                                                        |
| DCS-IR-TR-08 | Template Management Dashboard MUST provide lifecycle oversight of a... | <br>Partial | API: lifecycle oversight via search/status/history. UI out of harness.                                                                                                                                                                                                    |

### 3.1.1 UI — Contract Workflow (DCS-IR-CWE-...)

| ID            | Requirement                                                            | Status  | Evidence / disposition                                                                                                                                                                                                                                                               |
|---------------|------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-IR-CWE-01 | Contract Creation UI MUST allow Contract Creators to create and sub... | Partial | API: 03 create from approved template. UI out of harness.                                                                                                                                                                                                                            |
| DCS-IR-CWE-02 | Contract Creation UI MUST enable population of contract data, inclu... | Covered | API: parties/policies/evidence populated at create (03, 18, 05 evidence). UI now asserted: e2e/contract-fill.spec.ts fills a placeholder through the edit UI and the emitted document carries the forField-bound typed value in an odrl:Offer, with no editor-internal keys leaking. |
| DCS-IR-CWE-03 | Contract Negotiation UI MUST allow parties to exchange responses, r... | Partial | API: negotiation responses/redlines/comments (03). UI out of harness.                                                                                                                                                                                                                |
| DCS-IR-CWE-04 | Contract Negotiation UI MUST support comparison of contract version... | Partial | API: version history compare (03). UI out of harness.                                                                                                                                                                                                                                |
| DCS-IR-CWE-05 | Contract Review UI MUST allow Reviewers to retrieve, inspect, and v... | Covered | 03 state-machine invalid-transition + approval-chain scenarios (tagged @DCS-IR-CWE-05): review path enforced.                                                                                                                                                                        |
| DCS-IR-CWE-06 | Contract Review UI MUST allow Reviewers to respond with findings, r... | Covered | Review responses with findings/comments (tagged @DCS-IR-CWE-06 on the state-machine scenarios; approval comments in 03/contract_approval).                                                                                                                                           |
| DCS-IR-CWE-07 | Contract Review UI MUST provide search capabilities to locate contr... | Partial | API: contract search by state/metadata/parent (03 state-filtered search, 20 parent_did filter). UI out of harness.                                                                                                                                                                   |
| DCS-IR-CWE-08 | Contract Approval UI MUST allow Approvers to retrieve contracts in ... | Partial | API: approvers retrieve reviewed contracts (03/contract_approval). UI out of harness.                                                                                                                                                                                                |
| DCS-IR-CWE-09 | Contract Approval UI MUST allow Approvers to approve, reject (with ... | Covered | Approve / reject-with-reason / resubmit proven (03/contract_approval + state machine).                                                                                                                                                                                               |
| DCS-IR-       | Contract Approval UI MUST ensure                                       | Covered | Approved contracts proceed to signing (03 approval-transition + 22, tagged @DCS-IR-CWE-10). Catalogue forwarding is a deliberate MANUAL user action by architectural decision: catalogue registration                                                                                |

| ID            | Requirement                                                            | Status                                                                                       | Evidence / disposition                                                                                                                                                                                  |
|---------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CWE-10        | approved contracts are forwarded i...                                  |                                                                                              | can fail, be re-run, or be unconfigured — an explicit action models that honestly (same rationale as template publication, 02/template_catalogue).                                                      |
| DCS-IR-CWE-11 | Contract Management Dashboard UI MUST allow Managers to retrieve an... | <br>Covered | API: lifecycle-wide search (03 state-filtered search). Dashboard UI now asserted: e2e/dashboards.spec.ts (contract dashboard lists contracts with their lifecycle state for the Contract Manager role). |
| DCS-IR-CWE-12 | Contract Management Dashboard UI MUST allow Managers to store evide... | <br>Partial | API: evidence store (05 TSA receipt), terminate (06), audits (08). UI out of harness.                                                                                                                   |
| DCS-IR-CWE-13 | Contract Management Dashboard UI MUST provide lifecycle monitoring ... | <br>Partial | API: lifecycle monitoring via states/history/KPIs (05). UI out of harness.                                                                                                                              |

### 3.1.1 UI — Storage & Archive (DCS-IR-CSA-...)

| ID            | Requirement                                                            | Status                                                                                         | Evidence / disposition                                                                                                                                                                                           |
|---------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-IR-CSA-01 | Archive Manager Dashboard UI MUST allow Archive Managers to retriev... | <br>Covered   | 07 retrieve+search archive scenarios (tagged @DCS-IR-CSA-01).                                                                                                                                                    |
| DCS-IR-CSA-02 | Archive Manager Dashboard UI MUST allow storing new contracts and e... | <br>Covered   | Evidence store into archive (05 TSA receipt); signed contracts auto-stored (05 archive-at-SIGNED).                                                                                                               |
| DCS-IR-CSA-03 | Archive Manager Dashboard UI MUST allow terminating or deleting arc... | <br>Covered   | Terminate covered (06); archive delete scenarios (07, tagged @DCS-FR-CSA-17).                                                                                                                                    |
| DCS-IR-CSA-04 | Archive Manager Dashboard UI MUST allow running audits on archive o... | <br>Covered   | Archive audit endpoint covered (07, tagged @DCS-IR-CSA-04).                                                                                                                                                      |
| DCS-IR-CSA-05 | Archive Access UI MUST allow Observers to retrieve and search archi... | <br>Covered  | 07 least-privilege access enforcement (tagged @DCS-IR-CSA-05).                                                                                                                                                   |
| DCS-IR-CSA-06 | Archive Access UI MUST ensure that read-only users cannot modify, t... | <br>Covered | 07 read-only Observer scenario: Contract Observer retrieves the archive (200) yet delete is denied — matches the design scoping (retrieve/search: Archive Manager+Observer; store/delete: Archive Manager only). |

### 3.1.1 UI — Signature Management (DCS-IR-SM-...)

| ID           | Requirement                                                            | Status                                                                                      | Evidence / disposition                                                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-IR-SM-01 | Secure Contract Viewer UI MUST allow Signers and Managers to retrie... |  Covered   | API: approved-contract retrieval for signing (22). Viewer UI now asserted: e2e/dashboards.spec.ts renders the human-readable document from the machine-readable JSON-LD in the contract view (Contract Content tab).                                  |
| DCS-IR-SM-02 | Secure Contract Viewer UI MUST allow verification of contract integ... |  Covered   | Integrity/envelope verification via verify endpoints (08, 19, 22 verify cross-check).                                                                                                                                                                 |
| DCS-IR-SM-03 | Secure Contract Viewer UI MUST allow applying signatures with appro... |  Covered   | Signature application with verified credentials (22 ceremony-gate + webhook/PID scenarios).                                                                                                                                                           |
| DCS-IR-SM-04 | Secure Contract Viewer UI MUST allow validation of applied signatur... |  Covered   | Applied-signature validation endpoint scenario (04, tagged @DCS-FR-SM-18).                                                                                                                                                                            |
| DCS-IR-SM-05 | Signature Compliance Viewer UI MUST allow compliance users to valid... |  Partial   | Compliance users (Compliance Officer/Auditor scopes) read GET /signature/view (04) with cryptographic integrity findings from the shared validation machinery; trust anchors/proofs/timestamps validated in verify paths (21, 22). UI out of harness. |
| DCS-IR-SM-06 | Signature Compliance Viewer UI MUST allow revocation of signatures ... |  Covered | 15 signature revocation (tagged @DCS-FR-SM-20).                                                                                                                                                                                                       |
| DCS-IR-SM-07 | Signature Compliance Viewer UI MUST allow running compliance checks... |  Covered | Compliance-check endpoint scenario (04, tagged @DCS-FR-SM-21).                                                                                                                                                                                        |
| DCS-IR-SM-08 | Signature Compliance Viewer UI MUST allow generating audit reports ... |  Covered | Signature audit-report scenario (04, tagged @DCS-FR-SM-19).                                                                                                                                                                                           |



### 3.1.1 UI — Process Audit & Compliance (DCS-IR-PACM-...)

| ID             | Requirement                                                            | Status                                                                                       | Evidence / disposition                                                                                                                                |
|----------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-IR-PACM-01 | Auditing Tool UI MUST allow Auditors to initiate audits across cont... | <br>Covered | 08 scoped-audit scenario (tagged @DCS-IR-PACM-01).                                                                                                    |
| DCS-IR-PACM-02 | Auditing Tool UI MUST provide reporting capabilities with exportabl... | <br>Covered | 08 report-generation scenario (tagged @DCS-IR-PACM-02).                                                                                               |
| DCS-IR-PACM-03 | Non-Compliance Investigation UI MUST allow Compliance Officers to c... | <br>Covered | 08 continuous monitoring with structured checked_at+risks response; risk detection during approval incl. PAC-trail anchoring in 03/contract_approval. |
| DCS-IR-PACM-04 | Non-Compliance Investigation UI MUST allow incident reporting and l... | <br>Covered | 08 incident-reporting scenario (tagged @DCS-IR-PACM-04).                                                                                              |

### 3.1.2 Hardware Interfaces (DCS-IR-HI-...)

| ID           | Requirement                                         | Status                                                                                           | Evidence / disposition                                                                                                                          |
|--------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-IR-HI-01 | Interface for Use of Signing Secrets (HSM/QSCD/TPM) | <br>Covered   | 21 pack: PKCS#11/SoftHSM-backed keys, ES256 everywhere, rotation + CRL (tagged @DCS-IR-HI-01).                                                  |
| DCS-IR-HI-02 | FIDO2 Security Key Interface                        | <br>Deviation | FIDO2/WebAuthn login not implemented (no WebAuthn endpoints). Hardware-authenticator flows also not automatable headlessly. Recorded deviation. |
| DCS-IR-HI-03 | Platform TPM 2.0 / Secure Enclave Interface         | <br>Deviation | TPM sealing/remote attestation not implemented; platform-infra concern. Recorded deviation.                                                     |

### 3.4 Business Rules (DCS-NFR-BR-...)

| ID            | Requirement                            | Status                                                                                      | Evidence / disposition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|----------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-NFR-BR-01 | Strong Authentication & Role Binding   |  Covered   | Wallet-VC login is the only auth path — there is no password or non-VP fallback (see SI-08); roles bind from the verified presentation and RBAC is enforced everywhere (01). The second factor (wallet unlock / holder authentication) lives in the wallet, outside the DCS boundary.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DCS-NFR-BR-02 | Participant Eligibility                |  Covered   | 17: unverified/untrusted peers rejected on every DCS-to-DCS surface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| DCS-NFR-BR-03 | Legally Valid Signatures               |  Covered   | 05 non-SIGNED deploy refusal (tagged @DCS-NFR-BR-03); AES default (22).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| DCS-NFR-BR-04 | Template Governance                    |  Covered   | Contract create only from REGISTERED templates (03 steps + 02 approval chain).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| DCS-NFR-BR-05 | Immutable Auditability                 |  Covered  | Hash-chained TSA/IPFS audit for all lifecycle actions (08) + RBAC on logs (08 non-auditor denial).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| DCS-NFR-BR-06 | Revocation & Termination Propagation   |  Covered | Signature revocation → REVOKED immediately (15); cross-instance propagation via the synchronizer's SignatureManagement broadcast (17 revocation-propagation scenario: revoke on A, REVOKED replicated on B through the JAdES-verified post_sync path).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| DCS-NFR-BR-07 | Token & API Control                    |  Partial | Role-scoped tokens enforced (01); explicit minimal-scope token issuance policy is IdP config — noted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DCS-NFR-BR-08 | DCS-to-DCS Interoperability Safeguards |  Covered | 17 pack (tagged @NFR-BR-08): authenticated, trusted-peer-only exchanges with audit. Phase 4: post_sync (backend/internal/service/dcs_to_dcs.go) calls validation.RemoteShapeSource/VerifyAgainstOriginatorHub after the four existing trust layers accept a synced contract — it resolves the document's sh:shapesGraph anchor back to the ORIGINATOR's public Semantic Hub and re-validates against those exact shapes, not the receiver's own local hub (best-effort/non-blocking: a peer hub outage never fails an otherwise-trusted sync). 17/two_instance_peer_trust "A contract synced from instance A carries a sh:shapesGraph anchor resolvable against instance A's own Semantic Hub" (@two-instance) proves the reachability precondition end to end; the validation logic itself is proven by TestVerifyAgainstOriginatorHub (httpstest-simulated peer hub). |

| ID            | Requirement                  | Status                                                                                       | Evidence / disposition                                      |
|---------------|------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| DCS-NFR-BR-09 | Catalogue-Aligned Publishing | <br>Covered | Catalogue publish/consume scenario (02/template_catalogue). |

### 3.3.3 Security (DCS-NFR-SEC-...)

| ID             | Requirement                            | Status                                                                                                 | Evidence / disposition                                                                                                                                                                                                                        |
|----------------|----------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS-NFR-SEC-01 | Transport Layer Security               |  Not BDD-verifiable   | TLS 1.3-only is deployment/ingress config; not asserted from the plaintext BDD env.                                                                                                                                                           |
| DCS-NFR-SEC-02 | State-of-the-art Cryptography          |  Covered              | 21: P-256/ES256 across DID/JAR/VC/C2PA/PAdES, no legacy RSA (tagged @DCS-NFR-SEC-02).                                                                                                                                                         |
| DCS-NFR-SEC-03 | Authentication and Authorization       |  Covered              | 01 pack + role negatives suite-wide; party read-scoping on retrieve_by_id (03: dcs:parties gate, 403 forbidden, CONTRACT_ACCESS_DENIED audit event; Sys.*/Auditor org-independent; peer-adopted contracts readable by the adopting instance). |
| DCS-NFR-SEC-04 | Integrity Protection for Configuration |  Not BDD-verifiable   | Config integrity (signed/authenticated config) is platform concern; Helm-managed config — review-verified.                                                                                                                                    |
| DCS-NFR-SEC-05 | Integrity Protection for Service       |  Not BDD-verifiable   | Service integrity/attestation — platform concern (image digests, admission control).                                                                                                                                                          |
| DCS-NFR-SEC-06 | Storage of Secrets                     |  Covered             | Private keys live in PKCS#11 token only (21 DID-key scenarios; provisioning scripts).                                                                                                                                                         |
| DCS-NFR-SEC-07 | Testing                                |  Not BDD-verifiable | Process requirement — this BDD suite + Go tests + linters + CI are the evidence; pentest is external.                                                                                                                                         |
| DCS-NFR-SEC-08 | Confidentiality                        |  Partial            | RBAC + party read-scoping proven at API level (03 party-access scenarios, 403 + audit trail); storage-level encryption is infra (SEC-14).                                                                                                     |
| DCS-NFR-SEC-09 | Monitoring, Logging & Auditability     |  Covered            | Immutable audit logs retrievable for audits (08); /metrics exposed (16/prometheus).                                                                                                                                                           |
| DCS-NFR-SEC-10 | Data Integrity                         |  Covered            | Per-resource hash chains + Merkle checkpoint roots (ADR-16) + tamper-detection scenarios (03/format_review) + C2PA/PAdES integrity (19/22).                                                                                                   |
| DCS-NFR-SEC-11 | Monitoring & Incident Response         |  Partial            | Prometheus /metrics (16); automated incident response not modeled — noted.                                                                                                                                                                    |
| DCS-NFR-SEC-12 | Secure Configuration Management        |  Not BDD-verifiable | Secure config management — GitOps/platform concern.                                                                                                                                                                                           |
| DCS-NFR-       | Secure Data Disposal                   |  Partial            | Archive delete with audit (07); cryptographic erasure policy is infra — noted.                                                                                                                                                                |

| ID             | Requirement                                  | Status                                                                                               | Evidence / disposition                                                                                           |
|----------------|----------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| SEC-13         |                                              |                                                                                                      |                                                                                                                  |
| DCS-NFR-SEC-14 | Data Encryption at Rest & In Transit         |  Not BDD-verifiable | Encryption at rest — storage/platform config; in transit see SEC-01.                                             |
| DCS-NFR-SEC-15 | Secure Software Development Lifecycle (SDLC) |  Not BDD-verifiable | SDLC process — lint/hooks/CI in repo; review-verified.                                                           |
| DCS-NFR-SEC-16 | Identity Federation                          |  Partial            | OIDC federation via Hydra proven; third-party IdP interop is config — noted.                                     |
| DCS-NFR-SEC-17 | Secure Boot & Hardware Security              |  Deviation          | Secure boot — out of software scope for this service; platform deviation.                                        |
| DCS-NFR-SEC-18 | Selective Disclosure for Privacy             |  Covered            | SD-JWT selective disclosure in ceremony webhook (22 webhook + verbatim-embedding scenarios, tagged @NFR-SEC-18). |

### 3.3.2 Safety (DCS-NFR-SF-...)

| ID            | Requirement                             | Status                                                                                                 | Evidence / disposition                                                                                          |
|---------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| DCS-NFR-SF-01 | Reset Possibility                       |  Not BDD-verifiable | Stateless pods restarted throughout suite runs (rollouts during deploys) — k8s-level property; review-verified. |
| DCS-NFR-SF-02 | Remote Administration                   |  Not BDD-verifiable | Remote administration channel — operations concern.                                                             |
| DCS-NFR-SF-03 | Business Continuity & Disaster Recovery |  Not BDD-verifiable | BC/DR — operations concern (RTO/RPO).                                                                           |

### 3.3.1 Performance (DCS-NFR-PER-...)

| ID             | Requirement               | Status                                                                                                 | Evidence / disposition                                                                                |
|----------------|---------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| DCS-NFR-PER-01 | Performance by Design     |  Not BDD-verifiable | Performance-by-design — evidenced anecdotally by suite runtime; formal load testing out of BDD scope. |
| DCS-NFR-PER-02 | Scalability               |  Not BDD-verifiable | Scalability — load testing out of BDD scope.                                                          |
| DCS-NFR-PER-03 | Availability & Resilience |  Not BDD-verifiable | Availability/resilience — ops concern; k8s restarts exercised incidentally.                           |

### 3.3.4 Software Quality (DCS-NFR-SQ-...)

| ID            | Requirement               | Status                                                                                               | Evidence / disposition                                                                                    |
|---------------|---------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| DCS-NFR-SQ-01 | Programming Style         |  Not BDD-verifiable | Code style — golangci-lint/ESLint/Prettier + hooks; review-verified.                                      |
| DCS-NFR-SQ-02 | Build Scripts             |  Covered            | Build scripts exist and are exercised by CI + dev-stack (repo Makefiles/scripts) — process evidence.      |
| DCS-NFR-SQ-03 | Containerized Deployment  |  Covered            | Docker multi-stage + Helm on kind is exactly how this suite runs — continuously proven in CI.             |
| DCS-NFR-SQ-04 | Privacy by Design         |  Not BDD-verifiable | Privacy by design — process; selective disclosure covered (SEC-18).                                       |
| DCS-NFR-SQ-05 | Non-Repudiation           |  Covered            | Non-repudiation: PAdES + RFC3161 TSA + signer identity + immutable audit (22, 05 TSA receipt).            |
| DCS-NFR-SQ-06 | System Interoperability   |  Partial            | Interoperability: DCS-to-DCS (17), ORCE (05 round-trip), OIDC; broader enterprise-system matrix untested. |
| DCS-NFR-SQ-07 | Usability & Accessibility |  Not BDD-verifiable | WCAG/usability — UI concern out of harness.                                                               |
| DCS-NFR-SQ-08 | Orchestration Layer       |  Covered           | FACIS/XFSC ORCE integration is load-bearing in the suite (TSA + contract-target flows).                   |

### 3.3.5 Compliance (DCS-NFR-COMP-...)

| ID              | Requirement      | Status                                                                                                 | Evidence / disposition                                                                                      |
|-----------------|------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| DCS-NFR-COMP-01 | Legal Compliance |  Not BDD-verifiable | Legal/GDPR/eIDAS compliance — documentation/process requirement (includes COMP-02/03 embedded in SRS text). |

## Use-case map (SRS §4 / [DCS-FR-UC-XX-Y] linkage sections)

| UC    | Title                                 | Feature pack(s)                                                                                                                                                               |
|-------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UC-01 | User Authentication & Authorization   | 01_authentication_authorization                                                                                                                                               |
| UC-02 | Contract Template Management          | 02_template_management (incl. catalogue)                                                                                                                                      |
| UC-03 | Contract Creation                     | 03_contract_creation (creation, negotiation, approval, format review, state machine)                                                                                          |
| UC-04 | Contract Signing                      | 22_real_signing_vertical, 04_contract_signing, 21_pki_consolidation                                                                                                           |
| UC-05 | Contract Deployment                   | 05_contract_deployment                                                                                                                                                        |
| UC-06 | Contract Lifecycle Management         | 06_contract_lifecycle (termination + renewal), 19 (expiry)                                                                                                                    |
| UC-07 | Contract Storage & Security           | 07_contract_storage_security, 20 (bundles/audit)                                                                                                                              |
| UC-08 | Contract Compliance & Auditing        | 08_audit_compliance, 18_odrl_soundness                                                                                                                                        |
| UC-09 | DCS Administration                    | RBAC config is IdP/Helm config (  ); role enforcement covered by 01 + negatives suite-wide |
| UC-10 | Contract Automation & Integration     | 05 (ORCE), 12 (API automation), 18 (integrity gates)                                                                                                                          |
| UC-11 | API & System Integrations             | 05, 12, 17; catalogue (02)                                                                                                                                                    |
| UC-12 | System-Based Contract Management      | 12_system_based_contract_management (contract_lifecycle_via_api, system_user_classes)                                                                                         |
| UC-13 | External System Contract Execution    | 05_contract_deployment (target-system deploy/callback/evidence)                                                                                                               |
| UC-14 | Identity & PoA Credential Acquisition | 22 (PID identity); PoA = deviation (14_credential_acquisition documents it)                                                                                                   |
| UC-15 | Access Rights Revocation              | 15_access_revocation, 21 CRL revocation                                                                                                                                       |

## Deviation register (capabilities the product does not implement — honest , no fake scenarios)

| Item                                    | Requirement(s)               | Note                                                                                                                              |
|-----------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| QES execution                           | DCS-FR-SM-01                 | Needs qualified TSP/QSCD; AES delivered                                                                                           |
| PoA credential acquisition + chain-walk | DCS-FR-SM-03/04, UC-14       | PoA presented at login with status checking; issuer chain-walk is deferred roadmap work — 14 pack keeps tagged @skip placeholders |
| Configurable expiry/alert notifications | DCS-FR-CSA-04/20             | Detection covered; delivery channels absent                                                                                       |
| FIDO2/WebAuthn                          | DCS-IR-HI-02                 | No WebAuthn endpoints                                                                                                             |
| TPM sealing / remote attestation        | DCS-IR-HI-03, DCS-NFR-SEC-17 | Platform concern, not implemented                                                                                                 |
| "Replaced" C2PA lifecycle banner        | (19 lifecycle-banner subset) | Explicit scope decision, tracked in 19 pack header                                                                                |

OID4VP-as-login (DCS-IR-SI-08, DCS-NFR-BR-01) is exercised by every scenario's authentication: the harness performs the full headless VP login (login → challenge binding → JAR → vp\_token → verification → token exchange) and the backend accepts a Hydra login only through that path. A dedicated rejected-presentation-at-login negative remains follow-up work, not a deviation. Multi-signer flows (DCS-FR-SM-07/17) graduated from this list: 22/multi\_signer asserts them end-to-end, including two distinct signer identities and the deploy gate.

## Semantic canonicity evidence (interoperability assertion)

Claim: every artifact the DCS produces is standard, canonical semantics — consumable by off-the-shelf JSON-LD/RDF/ODRL/SHACL tooling with no DCS-specific conventions beyond a served, declared ODRL profile.



| Assertion                                            | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documents are valid JSON-LD 1.1; every term resolves | <code>@context</code> is the Semantic Hub's versioned, dereferenceable URL (23/semantic_hub resolve scenarios); json-gold expansion round-trips in every audit ( <code>expandForAudit</code> , backend/internal/base/validation/odrlexpanded.go); external contexts resolve or normalization fails ( <code>validateExternalContextsResolvable</code> ).                                                                           |
| Policy sets are conformant ODRL 2.2                  | One enclosing <code>odrl:Offer</code> (unsigned) / <code>odrl:Agreement</code> (sealed at first signature — 18 "first signature seals" scenario); rules under permission/prohibition/obligation only (odrl:duty bucket rejected); policy identity is <code>@id</code> (separate <code>uid</code> rejected); exactly one action + assigner/assignee/target per rule (18 structure scenarios; <code>validateODRLPolicySet</code> ). |
| Custom semantics are declared, not implied           | The DCS ODRL profile is a served document ( <code>/semantic/ontology/dcs-odrl-profile</code> ; docs/semantic-ontology/odrl/dcs-odrl-profile.ttl) declaring <code>dcs:provideCompliantValue</code> <code>odrl:includedIn</code> <code>odrl:use</code> and <code>dcs:RequirementField</code> <code>⊆</code> <code>odrl:LeftOperand</code> ; every policy set declares <code>odrl:profile</code> .                                   |
| Constraint↔value binding is plain graph traversal    | A submitted value references its field by IRI ( <code>dcs:forField</code> = the constraint's <code>odrl:leftOperand</code> ); enforced by SHACL ( <code>dcs:SemanticConditionValueShape</code> ) and exercised by every 18/05 enforcement scenario plus the Playwright fill spec (e2e/contract-fill.spec.ts).                                                                                                                     |
| Machine rules are prose-backed                       | <code>dcs:prose</code> required on every rule (Go gate + <code>Odrl*ProseShape</code> SHACL shapes); the Playwright builder spec asserts the emitted rule's prose dereferences to a document block.                                                                                                                                                                                                                               |
| SHACL validation is real SHACL                       | goRDFlib engine (ADR-9) against hub-served shapes pinned by <code>sh:shapesGraph</code> (ADR-8); version-pinning proven by 23 "stricter shapes version" scenario; shacl-form renders forms from the same raw Turtle (Gaia-X shapes render unmodified).                                                                                                                                                                            |
| Provenance is PROV-aligned                           | <code>dcs:derivedFromTemplate</code> <code>⊆</code> <code>prov:wasDerivedFrom</code> , <code>dcs:renewsContract</code> <code>⊆</code> <code>prov:wasRevisionOf</code> (generated OWL, docs/semantic-ontology/linkml); derivation carries the template <code>@id</code> + version as a node, not an opaque blob.                                                                                                                   |
| Canonical hashing is a standard                      | RFC 8785 JCS for the JADES payload and deployment content hash — byte-identical across Go ( <code>gowebpki/jcs</code> ) and Python ( <code>jcs</code> ), proven by the cross-language fixture test and 17/05 hash-verification scenarios.                                                                                                                                                                                         |
| Vocabulary identity is shared, not per-instance      | w3id.org namespaces are the vocabulary identity (hub serves the content); two federated instances speak the same terms (17 peer scenarios exchange and verify full documents). Public w3id dereferencing is the one external residual (registration PR).                                                                                                                                                                          |

## eIDAS 2.0 signature soundness evidence

Claim: the signature chain is architecturally sound under eIDAS 2.0 for advanced electronic signatures (AdES), with qualified-level (QES) execution being the single recorded deviation (no qualified TSP/QSCD reachable from the hermetic environment; `credential_type` carries the level end-to-end for when one is wired).

| Assertion                                           | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signer identification via the EUDI-wallet mechanism | Signing requires a completed OID4VP PID presentation ceremony (SD-JWT VC + KB-JWT), verified before any signature ( <code>ErrCeremonyRequired</code> , 22 ceremony-gate scenarios); the presentation is embedded verbatim into the signed PDF inside a signing-summary VC (embed-first-sign-second, 22 PID-embedding + verify cross-check).                                                                                                     |
| Signature format is a recognized AdES baseline      | PAdES signatures over PDF/A-3 via the HSM-backed P-256 key (PKCS#11/SoftHSM2, ADR-1; 21 internal-signing + CRL scenarios); DCS-to-DCS transport signatures are JAdES baseline-B (ETSI TS 119 182-1: sigT marked crit, x5c chain — backend/internal/base/jades, 17 peer verification scenarios).                                                                                                                                                 |
| Trust anchoring against the EU list                 | <code>EUTrustPool</code> (LOTL/TSL) gates every contract mutation and both peer-sync legs ( <code>VerifyEIDASCertificate</code> in <code>contract_workflow_engine.go</code> and <code>dcx_to_dcx.go</code> ); dev CA is a swappable TrustAnchor, not a bypass.                                                                                                                                                                                  |
| Revocation is enforced                              | Credential status lists checked on every verification via the mechanism-detecting status verifier (W3C BitstringStatusList/StatusList2021, IETF Token Status List over JOSE/COSE-CWT, XFSC — backend/internal/auth/oid4vp/status + <code>statuslist_verify.go</code> , <code>docs/status-list-verification.md</code> ); CRL revocation flips signing to rejection (21 CRL scenario); signature revocation is first-class (04 revoke scenarios). |
| Time evidence is RFC 3161                           | Archive entries and deployment receipts carry TSA timestamps verified against the provisioned TSA certificate (05 TSA-evidence scenario, 07 archive entries); JAdES carries claimed signing time (sigT).                                                                                                                                                                                                                                        |
| The signed bytes are the semantic document          | The content hash (RFC 8785) and the PAdES signature are computed over the sealed odr!Agreement document inside the signing transaction ( <code>apply.go</code> seal-before-hash); export/verify recompute and compare (03 <code>format_review</code> tamper seams, 20 export refusal).                                                                                                                                                          |
| Integrity of the archived artifact                  | C2PA manifest with lifecycle assertions stamped before signing (update-then-sign, ADR-4; 19 conformance scenarios), archive chain hash-linked and TSA-timestamped with signing evidence embedded (08 <code>audit_ui_archive_orce</code> feature).                                                                                                                                                                                               |
| What would make it QES                              | A qualified TSP signing certificate on a QSCD replacing the dev CA in the same PKCS#11 slot, plus a qualified TSA endpoint in <code>TSA_URL</code> — both configuration-level swaps by design (ADR-1, ADR-3); recorded as the DCS-FR-SM-01 partial.                                                                                                                                                                                             |